T3 group factors: EFT_1_after_F

Active theory: SM + S_1 + S_2 + $C_{LLS_1S_2}$. The fermion F is absent, so y_1, y_2 and M_F are not running EFT couplings.

$$j = \frac{d-1}{2}, \qquad C_2(d) = \frac{d^2-1}{4}, \qquad T(d) = \frac{d(d^2-1)}{12}.$$

In every comparison table, $\mathcal{S}$ denotes the RGE structure multiplying the displayed group factor in $16\pi^2\beta$.

Notation key

symbol	interaction / definition
g_1, g_2, g_3	SM gauge interactions
Y_u, Y_d, Y_e	SM Yukawa interactions
y_1	LFS_1 + h.c.
y_2	LFS_2 + h.c.
M_F	$-\bar{F}M_F F$
T	$\mathrm{Tr}(Y_e^\dagger Y_e + 3Y_u^\dagger Y_u + 3Y_d^\dagger Y_d)$
$T_\nu^{(1)}$	$\mathrm{Tr}(y_1^\dagger y_1)$
$T_\nu^{(2)}$	$\mathrm{Tr}(y_2^\dagger y_2)$
T_ν	$T_\nu^{(1)} + T_\nu^{(2)}$
m_1^2, m_2^2	$m_1^2 S_1^\dagger S_1,\ m_2^2 S_2^\dagger S_2$
λ_1	$\frac{1}{2}\lambda_1 (H^\dagger H)^2$
$\lambda_{S_i}^{(1)}$	$\frac{1}{2}\lambda_{S_i}^{(1)} (S_i^\dagger S_i)^2$
$\lambda_{HS_i}^{(1)}$	$(H^\dagger H)(S_i^\dagger S_i)$
$\lambda_{HS_i}^{(A)}$	$(H^\dagger T^A H)(S_i^\dagger T^A S_i)$
λ_5	$HHS_1S_2^\dagger$ + h.c.
$\lambda_{12}^{(1)}$	$(S_1^\dagger S_1)(S_2^\dagger S_2)$
$\lambda_{12}^{(A)}$	$(S_1^\dagger T^A S_1)(S_2^\dagger T^A S_2)$
$\lambda_{12}^{(\times)}$	$(S_1^\dagger S_2)(S_2^\dagger S_1)$

Stage field content

run	S_1	S_2	F	integrated at this stage	active after matching
T3-A-m4	$(1,-2)$	$(3,-1)$	$(2,-\frac{3}{2})$	F	S1, S2
T3-A-m2	$(1,-1)$	$(3,0)$	$(2,-\frac{1}{2})$	F	S1, S2
T3-A-p0	$(1,0)$	$(3,1)$	$(2,\frac{1}{2})$	F	S1, S2
T3-B-m3	$(2,-\frac{3}{2})$	$(2,-\frac{1}{2})$	$(1,-1)$	F	S1, S2
T3-B-m1	$(2,-\frac{1}{2})$	$(2,\frac{1}{2})$	$(1,0)$	F	S1, S2
T3-B-p1	$(2,\frac{1}{2})$	$(2,\frac{3}{2})$	$(1,1)$	F	S1, S2
T3-C-m3	$(2,-\frac{3}{2})$	$(2,-\frac{1}{2})$	$(3,-1)$	F	S1, S2
T3-C-m1	$(2,-\frac{1}{2})$	$(2,\frac{1}{2})$	$(3,0)$	F	S1, S2
T3-C-p1	$(2,\frac{1}{2})$	$(2,\frac{3}{2})$	$(3,1)$	F	S1, S2
T3-D-m2	$(3,-1)$	$(1,0)$	$(2,-\frac{1}{2})$	F	S1, S2
T3-D-p0	$(3,0)$	$(1,1)$	$(2,\frac{1}{2})$	F	S1, S2
T3-D-p2	$(3,1)$	$(1,2)$	$(2,\frac{3}{2})$	F	S1, S2
T3-E-m4	$(3,-2)$	$(3,-1)$	$(2,-\frac{3}{2})$	F	S1, S2
T3-E-m2	$(3,-1)$	$(3,0)$	$(2,-\frac{1}{2})$	F	S1, S2
T3-E-p0	$(3,0)$	$(3,1)$	$(2,\frac{1}{2})$	F	S1, S2
T3-E-p2	$(3,1)$	$(3,2)$	$(2,\frac{3}{2})$	F	S1, S2

Field representations

run	S_1	S_2	α
T3 − A − m4	$(1,-2)$	$(3,-1)$	-4
T3 − A − m2	$(1,-1)$	$(3,0)$	-2
T3 − A − p0	$(1,0)$	$(3,1)$	0
T3 − B − m3	$(2,-\frac{3}{2})$	$(2,-\frac{1}{2})$	-3
T3 − B − m1	$(2,-\frac{1}{2})$	$(2,\frac{1}{2})$	-1
T3 − B − p1	$(2,\frac{1}{2})$	$(2,\frac{3}{2})$	1
T3 − C − m3	$(2,-\frac{3}{2})$	$(2,-\frac{1}{2})$	-3
T3 − C − m1	$(2,-\frac{1}{2})$	$(2,\frac{1}{2})$	-1
T3 − C − p1	$(2,\frac{1}{2})$	$(2,\frac{3}{2})$	1
T3 − D − m2	$(3,-1)$	$(1,0)$	-2
T3 − D − p0	$(3,0)$	$(1,1)$	0
T3 − D − p2	$(3,1)$	$(1,2)$	2
T3 − E − m4	$(3,-2)$	$(3,-1)$	-4
T3 − E − m2	$(3,-1)$	$(3,0)$	-2
T3 − E − p0	$(3,0)$	$(3,1)$	0
T3 − E − p2	$(3,1)$	$(3,2)$	2

Gauge and Yukawa beta functions in this EFT

$\beta^{(1)}_{g_1}$

$\mathcal{S}$	T3 − A − m4 ($\alpha=-4$)	T3 − A − m2 ($\alpha=-2$)	T3 − A − p0 ($\alpha=0$)	T3 − B −
g_1^3	$\frac{55}{6}$	$\frac{43}{6}$	$\frac{47}{6}$	

$\mathcal{S}$	T3 − B − p1 ($\alpha=1$)	T3 − C − m3 ($\alpha=-3$)	T3 − C − m1 ($\alpha=-1$)	T3 − C −
g_1^3	$\frac{17}{2}$	$\frac{17}{2}$	$\frac{43}{6}$	$\frac{1}{2}$

$\mathcal{S}$	T3 − D − p0 ($\alpha=0$)	T3 − D − p2 ($\alpha=2$)	T3 − E − m4 ($\alpha=-4$)	T3 − E − m2
g_1^3	$\frac{43}{6}$	$\frac{55}{6}$	$\frac{71}{6}$	$\frac{47}{6}$

$\mathcal{S}$	T3 − E − p2 ($\alpha=2$)
g_1^3	$\frac{71}{6}$

$\beta^{(1)}_{g_2}$

$\mathcal{S}$	T3 − A − m4 ($\alpha=-4$)	T3 − A − m2 ($\alpha=-2$)	T3 − A − p0 ($\alpha=0$)	T3 − B −
g_2^3	$-\frac{5}{2}$	$-\frac{5}{2}$	$-\frac{5}{2}$	$-$

$\mathcal{S}$	T3 − B − p1 ($\alpha=1$)	T3 − C − m3 ($\alpha=-3$)	T3 − C − m1 ($\alpha=-1$)	T3 − C −
g_2^3	$-\frac{17}{6}$	$-\frac{17}{6}$	$-\frac{17}{6}$	$-$

$\mathcal{S}$	T3 − D − p0 ($\alpha=0$)	T3 − D − p2 ($\alpha=2$)	T3 − E − m4 ($\alpha=-4$)	T3 − E − m2

g_2^3	$-\frac{5}{2}$	$-\frac{5}{2}$	$-\frac{11}{6}$	$-\frac{1}{6}$

$\mathcal{S}$	T3 − E − p2 ($\alpha = 2$)
g_2^3	$-\frac{11}{6}$

$\beta_{g_3}^{(1)}$

$\mathcal{S}$	T3 − A − m4 ($\alpha = -4$)	T3 − A − m2 ($\alpha = -2$)	T3 − A − p0 ($\alpha = 0$)	T3 − B −
g_3^3	-7	-7	-7	

$\mathcal{S}$	T3 − B − p1 ($\alpha = 1$)	T3 − C − m3 ($\alpha = -3$)	T3 − C − m1 ($\alpha = -1$)	T3 − C −
g_3^3	-7	-7	-7	$-$

$\mathcal{S}$	T3 − D − p0 ($\alpha = 0$)	T3 − D − p2 ($\alpha = 2$)	T3 − E − m4 ($\alpha = -4$)	T3 − E − m2
g_3^3	-7	-7	-7	-7

$\mathcal{S}$	T3 − E − p2 ($\alpha = 2$)
g_3^3	-7

$\beta_{Y_u}^{(1)}$

$\mathcal{S}$	T3 − A − m4 ($\alpha = -4$)	T3 − A − m2 ($\alpha = -2$)	T3 − A − p0 ($\alpha = 0$)	T3 − B −
$(Y_d(Y_d)^\dagger Y_u)_{ij}$	$-\frac{3}{2}$	$-\frac{3}{2}$	$-\frac{3}{2}$	
$(Y_u(Y_u)^\dagger Y_u)_{ij}$	$\frac{3}{2}$	$\frac{3}{2}$	$\frac{3}{2}$	
$g_2^2(Y_u)_{ij}$	$-\frac{9}{4}$	$-\frac{9}{4}$	$-\frac{9}{4}$	
$g_3^2(Y_u)_{ij}$	-8	-8	-8	
$g_1^2(Y_u)_{ij}$	$-\frac{17}{12}$	$-\frac{17}{12}$	$-\frac{17}{12}$	$-$
$T(Y_u)_{ij}$	1	1	1	

$\mathcal{S}$	T3 − B − p1 ($\alpha = 1$)	T3 − C − m3 ($\alpha = -3$)	T3 − C − m1 ($\alpha = -1$)	T3 − C −
$(Y_d(Y_d)^\dagger Y_u)_{ij}$	$-\frac{3}{2}$	$-\frac{3}{2}$	$-\frac{3}{2}$	$-$
$(Y_u(Y_u)^\dagger Y_u)_{ij}$	$\frac{3}{2}$	$\frac{3}{2}$	$\frac{3}{2}$	
$g_2^2(Y_u)_{ij}$	$-\frac{9}{4}$	$-\frac{9}{4}$	$-\frac{9}{4}$	$-$
$g_3^2(Y_u)_{ij}$	-8	-8	-8	$-$
$g_1^2(Y_u)_{ij}$	$-\frac{17}{12}$	$-\frac{17}{12}$	$-\frac{17}{12}$	$-$
$T(Y_u)_{ij}$	1	1	1	1

$\mathcal{S}$	T3 − D − p0 ($\alpha = 0$)	T3 − D − p2 ($\alpha = 2$)	T3 − E − m4 ($\alpha = -4$)	T3 − E − m2
$(Y_d(Y_d)^\dagger Y_u)_{ij}$	$-\frac{3}{2}$	$-\frac{3}{2}$	$-\frac{3}{2}$	$-\frac{3}{2}$
$(Y_u(Y_u)^\dagger Y_u)_{ij}$	$\frac{3}{2}$	$\frac{3}{2}$	$\frac{3}{2}$	$\frac{3}{2}$
$g_2^2(Y_u)_{ij}$	$-\frac{9}{4}$	$-\frac{9}{4}$	$-\frac{9}{4}$	$-\frac{9}{4}$
$g_3^2(Y_u)_{ij}$	-8	-8	-8	-8
$g_1^2(Y_u)_{ij}$	$-\frac{17}{12}$	$-\frac{17}{12}$	$-\frac{17}{12}$	$-\frac{17}{12}$

$T(Y_u)_{ij}$	1	1	1	1

$\mathcal{S}$	T3 − E − p2 ($\alpha = 2$)
$(Y_d(Y_d)^\dagger Y_u)_{ij}$	$-\frac{3}{2}$
$(Y_u(Y_u)^\dagger Y_u)_{ij}$	$\frac{3}{2}$
$g_2^2(Y_u)_{ij}$	$-\frac{9}{4}$
$g_3^2(Y_u)_{ij}$	-8
$g_1^2(Y_u)_{ij}$	$-\frac{17}{12}$
$T(Y_u)_{ij}$	1

$\beta^{(1)}_{Y_d}$

$\mathcal{S}$	T3 − A − m4 ($\alpha = -4$)	T3 − A − m2 ($\alpha = -2$)	T3 − A − p0 ($\alpha = 0$)	T3 − B −
$(Y_d(Y_d)^\dagger Y_d)_{ij}$	$\frac{3}{2}$	$\frac{3}{2}$	$\frac{3}{2}$	
$(Y_u(Y_u)^\dagger Y_d)_{ij}$	$-\frac{3}{2}$	$-\frac{3}{2}$	$-\frac{3}{2}$	
$g_2^2(Y_d)_{ij}$	$-\frac{9}{4}$	$-\frac{9}{4}$	$-\frac{9}{4}$	
$g_3^2(Y_d)_{ij}$	-8	-8	-8	
$g_1^2(Y_d)_{ij}$	$-\frac{5}{12}$	$-\frac{5}{12}$	$-\frac{5}{12}$	
$T(Y_d)_{ij}$	1	1	1	

$\mathcal{S}$	T3 − B − p1 ($\alpha = 1$)	T3 − C − m3 ($\alpha = -3$)	T3 − C − m1 ($\alpha = -1$)	T3 − C −
$(Y_d(Y_d)^\dagger Y_d)_{ij}$	$\frac{3}{2}$	$\frac{3}{2}$	$\frac{3}{2}$	
$(Y_u(Y_u)^\dagger Y_d)_{ij}$	$-\frac{3}{2}$	$-\frac{3}{2}$	$-\frac{3}{2}$	
$g_2^2(Y_d)_{ij}$	$-\frac{9}{4}$	$-\frac{9}{4}$	$-\frac{9}{4}$	
$g_3^2(Y_d)_{ij}$	-8	-8	-8	
$g_1^2(Y_d)_{ij}$	$-\frac{5}{12}$	$-\frac{5}{12}$	$-\frac{5}{12}$	
$T(Y_d)_{ij}$	1	1	1	

$\mathcal{S}$	T3 − D − p0 ($\alpha = 0$)	T3 − D − p2 ($\alpha = 2$)	T3 − E − m4 ($\alpha = -4$)	T3 − E − m2
$(Y_d(Y_d)^\dagger Y_d)_{ij}$	$\frac{3}{2}$	$\frac{3}{2}$	$\frac{3}{2}$	$\frac{3}{2}$
$(Y_u(Y_u)^\dagger Y_d)_{ij}$	$-\frac{3}{2}$	$-\frac{3}{2}$	$-\frac{3}{2}$	$-\frac{3}{2}$
$g_2^2(Y_d)_{ij}$	$-\frac{9}{4}$	$-\frac{9}{4}$	$-\frac{9}{4}$	$-\frac{9}{4}$
$g_3^2(Y_d)_{ij}$	-8	-8	-8	-8
$g_1^2(Y_d)_{ij}$	$-\frac{5}{12}$	$-\frac{5}{12}$	$-\frac{5}{12}$	$-\frac{5}{12}$
$T(Y_d)_{ij}$	1	1	1	1

$\mathcal{S}$	T3 − E − p2 ($\alpha = 2$)
$(Y_d(Y_d)^\dagger Y_d)_{ij}$	$\frac{3}{2}$
$(Y_u(Y_u)^\dagger Y_d)_{ij}$	$-\frac{3}{2}$
$g_2^2(Y_d)_{ij}$	$-\frac{9}{4}$
$g_3^2(Y_d)_{ij}$	-8
$g_1^2(Y_d)_{ij}$	$-\frac{5}{12}$
$T(Y_d)_{ij}$	1

$\beta^{(1)}_{Y_e}$

$\mathcal{S}$	T3 − A − m4 ($\alpha=-4$)	T3 − A − m2 ($\alpha=-2$)	T3 − A − p0 ($\alpha=0$)	T3 − B −
$(Y_e(Y_e)^\dagger Y_e)_{ij}$	$\frac{3}{2}$	$\frac{3}{2}$	$\frac{3}{2}$	
$g_2^2(Y_e)_{ij}$	$-\frac{9}{4}$	$-\frac{9}{4}$	$-\frac{9}{4}$	
$g_1^2(Y_e)_{ij}$	$-\frac{15}{4}$	$-\frac{15}{4}$	$-\frac{15}{4}$	
$T(Y_e)_{ij}$	1	1	1	

$\mathcal{S}$	T3 − B − p1 ($\alpha=1$)	T3 − C − m3 ($\alpha=-3$)	T3 − C − m1 ($\alpha=-1$)	T3 − C −
$(Y_e(Y_e)^\dagger Y_e)_{ij}$	$\frac{3}{2}$	$\frac{3}{2}$	$\frac{3}{2}$	
$g_2^2(Y_e)_{ij}$	$-\frac{9}{4}$	$-\frac{9}{4}$	$-\frac{9}{4}$	
$g_1^2(Y_e)_{ij}$	$-\frac{15}{4}$	$-\frac{15}{4}$	$-\frac{15}{4}$	
$T(Y_e)_{ij}$	1	1	1	

$\mathcal{S}$	T3 − D − p0 ($\alpha=0$)	T3 − D − p2 ($\alpha=2$)	T3 − E − m4 ($\alpha=-4$)	T3 − E − m2
$(Y_e(Y_e)^\dagger Y_e)_{ij}$	$\frac{3}{2}$	$\frac{3}{2}$	$\frac{3}{2}$	
$g_2^2(Y_e)_{ij}$	$-\frac{9}{4}$	$-\frac{9}{4}$	$-\frac{9}{4}$	
$g_1^2(Y_e)_{ij}$	$-\frac{15}{4}$	$-\frac{15}{4}$	$-\frac{15}{4}$	
$T(Y_e)_{ij}$	1	1	1	

$\mathcal{S}$	T3 − E − p2 ($\alpha=2$)
$(Y_e(Y_e)^\dagger Y_e)_{ij}$	$\frac{3}{2}$
$g_2^2(Y_e)_{ij}$	$-\frac{9}{4}$
$g_1^2(Y_e)_{ij}$	$-\frac{15}{4}$
$T(Y_e)_{ij}$	1

$\beta_{m_1^2}$

Run block 1: T3-A-m4–T3-B-m1

$\mathcal{S}$	T3 − A − m4 ($\alpha=-4$)	T3 − A − m2 ($\alpha=-2$)	T3 − A − p0 ($\alpha=0$)	T3 − B − m3 ($\alpha=$
$\lambda_{S_1}^{(1)}\, m_1^2$	4	4	4	6
$g_2^2\, m_1^2$	0	0	0	$-\frac{9}{2}$
$g_1^2\, m_1^2$	−24	−6	0	$-\frac{27}{2}$
$\lambda_{12}^{(1)}\, m_2^2$	6	6	6	4
$\lambda_{S_1}^{(A)}\, m_1^2$	0	0	0	0
$\lambda_{12}^{(\times)}\, m_2^2$	0	0	0	0

Run block 2: T3-B-p1–T3-D-m2

$\mathcal{S}$	T3 − B − p1 ($\alpha=1$)	T3 − C − m3 ($\alpha=-3$)	T3 − C − m1 ($\alpha=-1$)	T3 − C − p1 ($\alpha=$
$\lambda_{S_1}^{(1)}\, m_1^2$	6	6	6	6
$g_2^2\, m_1^2$	$-\frac{9}{2}$	$-\frac{9}{2}$	$-\frac{9}{2}$	$-\frac{9}{2}$
$g_1^2\, m_1^2$	$-\frac{3}{2}$	$-\frac{27}{2}$	$-\frac{3}{2}$	$-\frac{3}{2}$
$\lambda_{12}^{(1)}\, m_2^2$	4	4	4	4
$\lambda_{S_1}^{(A)}\, m_1^2$	0	0	0	0
$\lambda_{12}^{(\times)}\, m_2^2$	0	0	0	0

Run block 3: T3-D-p0–T3-E-p0

$\mathcal{S}$	T3 − D − p0 ($\alpha = 0$)	T3 − D − p2 ($\alpha = 2$)	T3 − E − m4 ($\alpha = -4$)	T3 − E − m2 ($\alpha = -$
$\lambda_{S_1}^{(1)}\, m_1^2$	8	8	8	8
$g_2^2\, m_1^2$	−12	−12	−12	−12
$g_1^2\, m_1^2$	0	−6	−24	−6
$\lambda_{12}^{(1)}\, m_2^2$	2	2	6	6
$\lambda_{S_1}^{(A)}\, m_1^2$	4	4	4	4
$\lambda_{12}^{(\times)}\, m_2^2$	0	0	4	4

Run block 4: T3-E-p2–T3-E-p2

$\mathcal{S}$	T3 − E − p2 ($\alpha = 2$)
$\lambda_{S_1}^{(1)}\, m_1^2$	8
$g_2^2\, m_1^2$	−12
$g_1^2\, m_1^2$	−6
$\lambda_{12}^{(1)}\, m_2^2$	6
$\lambda_{S_1}^{(A)}\, m_1^2$	4
$\lambda_{12}^{(\times)}\, m_2^2$	4

$\beta_{m_2^2}$

Run block 1: T3-A-m4–T3-B-m1

$\mathcal{S}$	T3 − A − m4 ($\alpha = -4$)	T3 − A − m2 ($\alpha = -2$)	T3 − A − p0 ($\alpha = 0$)	T3 − B − m3 ($\alpha =$
$\lambda_{S_2}^{(1)}\, m_2^2$	8	8	8	6
$g_2^2\, m_2^2$	−12	−12	−12	$-\frac{9}{2}$
$g_1^2\, m_2^2$	−6	0	−6	$-\frac{3}{2}$
$\lambda_{12}^{(1)}\, m_1^2$	2	2	2	4
$\lambda_{S_2}^{(A)}\, m_2^2$	4	4	4	0
$\lambda_{12}^{(\times)}\, m_1^2$	0	0	0	0

Run block 2: T3-B-p1–T3-D-m2

$\mathcal{S}$	T3 − B − p1 ($\alpha = 1$)	T3 − C − m3 ($\alpha = -3$)	T3 − C − m1 ($\alpha = -1$)	T3 − C − p1 ($\alpha =$
$\lambda_{S_2}^{(1)}\, m_2^2$	6	6	6	6
$g_2^2\, m_2^2$	$-\frac{9}{2}$	$-\frac{9}{2}$	$-\frac{9}{2}$	$-\frac{9}{2}$
$g_1^2\, m_2^2$	$-\frac{27}{2}$	$-\frac{3}{2}$	$-\frac{3}{2}$	$-\frac{27}{2}$
$\lambda_{12}^{(1)}\, m_1^2$	4	4	4	4
$\lambda_{S_2}^{(A)}\, m_2^2$	0	0	0	0
$\lambda_{12}^{(\times)}\, m_1^2$	0	0	0	0

Run block 3: T3-D-p0–T3-E-p0

$\mathcal{S}$	T3 − D − p0 ($\alpha = 0$)	T3 − D − p2 ($\alpha = 2$)	T3 − E − m4 ($\alpha = -4$)	T3 − E − m2 ($\alpha = -$
$\lambda_{S_2}^{(1)}\, m_2^2$	4	4	8	8
$g_2^2\, m_2^2$	0	0	−12	−12
$g_1^2\, m_2^2$	−6	−24	−6	0
$\lambda_{12}^{(1)}\, m_1^2$	6	6	6	6
$\lambda_{S_2}^{(A)}\, m_2^2$	0	0	4	4
$\lambda_{12}^{(\times)}\, m_1^2$	0	0	4	4

Run block 4: T3-E-p2–T3-E-p2

$\mathcal{S}$	T3 − E − p2 ($\alpha = 2$)
$\lambda_{S_2}^{(1)}\, m_2^2$	8
$g_2^2\, m_2^2$	−12
$g_1^2\, m_2^2$	−24
$\lambda_{12}^{(1)}\, m_1^2$	6
$\lambda_{S_2}^{(A)}\, m_2^2$	4
$\lambda_{12}^{(\times)}\, m_1^2$	4

$\beta_{\lambda_{HS_1}^{(1)}}$

Run block 1: T3-A-m4–T3-B-m1

$\mathcal{S}$	T3 − A − m4 ($\alpha = -4$)	T3 − A − m2 ($\alpha = -2$)	T3 − A − p0 ($\alpha = 0$)	T3 − B − m3 ($\alpha =$
$\lvert\lambda_5\rvert^2$	12	12	12	6
$g_2^2\, \lambda_{HS_1}^{(1)}$	$-\frac{9}{2}$	$-\frac{9}{2}$	$-\frac{9}{2}$	−9
$g_1^2\, \lambda_{HS_1}^{(1)}$	$-\frac{51}{2}$	$-\frac{15}{2}$	$-\frac{3}{2}$	−15
$\lambda_1\, \lambda_{HS_1}^{(1)}$	6	6	6	6
$\left(\lambda_{HS_1}^{(1)}\right)^2$	4	4	4	4
$\lambda_{HS_1}^{(1)}\, \lambda_{S_1}^{(1)}$	4	4	4	6
$\lambda_{12}^{(1)}\, \lambda_{HS_2}^{(1)}$	6	6	6	4
g_2^4	0	0	0	$\frac{9}{4}$
g_1^4	12	3	0	$\frac{27}{4}$
$\lambda_{HS_1}^{(1)}\, T$	2	2	2	2
$\left(\lambda_{HS_1}^{(A)}\right)^2$	0	0	0	$\frac{3}{4}$
$\lambda_{HS_1}^{(1)}\, \lambda_{S_1}^{(A)}$	0	0	0	0
$\lambda_{HS_2}^{(1)}\, \lambda_{12}^{(\times)}$	0	0	0	0

Run block 2: T3-B-p1–T3-D-m2

$\mathcal{S}$	T3 − B − p1 ($\alpha = 1$)	T3 − C − m3 ($\alpha = -3$)	T3 − C − m1 ($\alpha = -1$)	T3 − C − p1 ($\alpha =$
$\lvert\lambda_5\rvert^2$	6	6	6	6
$g_2^2\, \lambda_{HS_1}^{(1)}$	−9	−9	−9	−9
$g_1^2\, \lambda_{HS_1}^{(1)}$	−3	−15	−3	−3
$\lambda_1\, \lambda_{HS_1}^{(1)}$	6	6	6	6
$\left(\lambda_{HS_1}^{(1)}\right)^2$	4	4	4	4
$\lambda_{HS_1}^{(1)}\, \lambda_{S_1}^{(1)}$	6	6	6	6
$\lambda_{12}^{(1)}\, \lambda_{HS_2}^{(1)}$	4	4	4	4
g_2^4	$\frac{9}{4}$	$\frac{9}{4}$	$\frac{9}{4}$	$\frac{9}{4}$
g_1^4	$\frac{3}{4}$	$\frac{27}{4}$	$\frac{3}{4}$	$\frac{3}{4}$
$\lambda_{HS_1}^{(1)}\, T$	2	2	2	2
$\left(\lambda_{HS_1}^{(A)}\right)^2$	$\frac{3}{4}$	$\frac{3}{4}$	$\frac{3}{4}$	$\frac{3}{4}$
$\lambda_{HS_1}^{(1)}\, \lambda_{S_1}^{(A)}$	0	0	0	0
$\lambda_{HS_2}^{(1)}\, \lambda_{12}^{(\times)}$	0	0	0	0

Run block 3: T3-D-p0–T3-E-p0

$\mathcal{S}$	T3 − D − p0 ($\alpha = 0$)	T3 − D − p2 ($\alpha = 2$)	T3 − E − m4 ($\alpha = -4$)	T3 − E − m2 ($\alpha = -$
$\lvert\lambda_5\rvert^2$	4	4	4	4
$g_2^2\, \lambda_{HS_1}^{(1)}$	$-\frac{33}{2}$	$-\frac{33}{2}$	$-\frac{33}{2}$	$-\frac{33}{2}$

$g_1^2\,\lambda_{HS_1}^{(1)}$	$-\frac{3}{2}$	$-\frac{15}{2}$	$-\frac{51}{2}$	$-\frac{15}{2}$
$\lambda_1\,\lambda_{HS_1}^{(1)}$	6	6	6	6
$\left(\lambda_{HS_1}^{(1)}\right)^2$	4	4	4	4
$\lambda_{HS_1}^{(1)}\,\lambda_{S_1}^{(1)}$	8	8	8	8
$\lambda_{12}^{(1)}\,\lambda_{HS_2}^{(1)}$	2	2	6	6
g_2^4	6	6	6	6
g_1^4	0	3	12	3
$\lambda_{HS_1}^{(1)}\,T$	2	2	2	2
$\left(\lambda_{HS_1}^{(A)}\right)^2$	2	2	2	2
$\lambda_{HS_1}^{(1)}\,\lambda_{S_1}^{(A)}$	4	4	4	4
$\lambda_{HS_2}^{(1)}\,\lambda_{12}^{(\times)}$	0	0	4	4

Run block 4: T3-E-p2–T3-E-p2

$\mathcal{S}$	$\mathrm{T3-E-p2}\ (\alpha=2)$
$\lvert\lambda_5\rvert^2$	4
$g_2^2\,\lambda_{HS_1}^{(1)}$	$-\frac{33}{2}$
$g_1^2\,\lambda_{HS_1}^{(1)}$	$-\frac{15}{2}$
$\lambda_1\,\lambda_{HS_1}^{(1)}$	6
$\left(\lambda_{HS_1}^{(1)}\right)^2$	4
$\lambda_{HS_1}^{(1)}\,\lambda_{S_1}^{(1)}$	8
$\lambda_{12}^{(1)}\,\lambda_{HS_2}^{(1)}$	6
g_2^4	6
g_1^4	3
$\lambda_{HS_1}^{(1)}\,T$	2
$\left(\lambda_{HS_1}^{(A)}\right)^2$	2
$\lambda_{HS_1}^{(1)}\,\lambda_{S_1}^{(A)}$	4
$\lambda_{HS_2}^{(1)}\,\lambda_{12}^{(\times)}$	4

$\beta_{\lambda_{HS_2}^{(1)}}$

Run block 1: T3-A-m4–T3-B-m1

$\mathcal{S}$	$\mathrm{T3-A-m4}\ (\alpha=-4)$	$\mathrm{T3-A-m2}\ (\alpha=-2)$	$\mathrm{T3-A-p0}\ (\alpha=0)$	$\mathrm{T3-B-m3}\ (\alpha=$
$\lvert\lambda_5\rvert^2$	4	4	4	6
$g_2^2\,\lambda_{HS_2}^{(1)}$	$-\frac{33}{2}$	$-\frac{33}{2}$	$-\frac{33}{2}$	-9
$g_1^2\,\lambda_{HS_2}^{(1)}$	$-\frac{15}{2}$	$-\frac{3}{2}$	$-\frac{15}{2}$	-3
$\lambda_1\,\lambda_{HS_2}^{(1)}$	6	6	6	6
$\left(\lambda_{HS_2}^{(1)}\right)^2$	4	4	4	4
$\lambda_{HS_2}^{(1)}\,\lambda_{S_2}^{(1)}$	8	8	8	6
$\lambda_{12}^{(1)}\,\lambda_{HS_1}^{(1)}$	2	2	2	4
g_2^4	6	6	6	$\frac{9}{4}$
g_1^4	3	0	3	$\frac{3}{4}$
$\left(\lambda_{HS_2}^{(A)}\right)^2$	2	2	2	$\frac{3}{4}$
$\lambda_{HS_2}^{(1)}\,\lambda_{S_2}^{(A)}$	4	4	4	0
$\lambda_{HS_2}^{(1)}\,T$	2	2	2	2
$\lambda_{HS_1}^{(1)}\,\lambda_{12}^{(\times)}$	0	0	0	0

Run block 2: T3-B-p1–T3-D-m2

$\mathcal{S}$	T3 − B − p1 ($\alpha = 1$)	T3 − C − m3 ($\alpha = -3$)	T3 − C − m1 ($\alpha = -1$)	T3 − C − p1 ($\alpha =$
$\lvert\lambda_5\rvert^2$	6	6	6	6
$g_2^2\,\lambda_{HS_2}^{(1)}$	-9	-9	-9	-9
$g_1^2\,\lambda_{HS_2}^{(1)}$	-15	-3	-3	-15
$\lambda_1\,\lambda_{HS_2}^{(1)}$	6	6	6	6
$\left(\lambda_{HS_2}^{(1)}\right)^2$	4	4	4	4
$\lambda_{HS_2}^{(1)}\,\lambda_{S_2}^{(1)}$	6	6	6	6
$\lambda_{12}^{(1)}\,\lambda_{HS_1}^{(1)}$	4	4	4	4
g_2^4	$\frac{9}{4}$	$\frac{9}{4}$	$\frac{9}{4}$	$\frac{9}{4}$
g_1^4	$\frac{27}{4}$	$\frac{3}{4}$	$\frac{3}{4}$	$\frac{27}{4}$
$\left(\lambda_{HS_2}^{(A)}\right)^2$	$\frac{3}{4}$	$\frac{3}{4}$	$\frac{3}{4}$	$\frac{3}{4}$
$\lambda_{HS_2}^{(1)}\,\lambda_{S_2}^{(A)}$	0	0	0	0
$\lambda_{HS_2}^{(1)}\,T$	2	2	2	2
$\lambda_{HS_1}^{(1)}\,\lambda_{12}^{(\times)}$	0	0	0	0

Run block 3: T3-D-p0–T3-E-p0

$\mathcal{S}$	T3 − D − p0 ($\alpha = 0$)	T3 − D − p2 ($\alpha = 2$)	T3 − E − m4 ($\alpha = -4$)	T3 − E − m2 ($\alpha = -$
$\lvert\lambda_5\rvert^2$	12	12	4	4
$g_2^2\,\lambda_{HS_2}^{(1)}$	$-\frac{9}{2}$	$-\frac{9}{2}$	$-\frac{33}{2}$	$-\frac{33}{2}$
$g_1^2\,\lambda_{HS_2}^{(1)}$	$-\frac{15}{2}$	$-\frac{51}{2}$	$-\frac{15}{2}$	$-\frac{3}{2}$
$\lambda_1\,\lambda_{HS_2}^{(1)}$	6	6	6	6
$\left(\lambda_{HS_2}^{(1)}\right)^2$	4	4	4	4
$\lambda_{HS_2}^{(1)}\,\lambda_{S_2}^{(1)}$	4	4	8	8
$\lambda_{12}^{(1)}\,\lambda_{HS_1}^{(1)}$	6	6	6	6
g_2^4	0	0	6	6
g_1^4	3	12	3	0
$\left(\lambda_{HS_2}^{(A)}\right)^2$	0	0	2	2
$\lambda_{HS_2}^{(1)}\,\lambda_{S_2}^{(A)}$	0	0	4	4
$\lambda_{HS_2}^{(1)}\,T$	2	2	2	2
$\lambda_{HS_1}^{(1)}\,\lambda_{12}^{(\times)}$	0	0	4	4

Run block 4: T3-E-p2–T3-E-p2

$\mathcal{S}$	T3 − E − p2 ($\alpha = 2$)
$\lvert\lambda_5\rvert^2$	4
$g_2^2\,\lambda_{HS_2}^{(1)}$	$-\frac{33}{2}$
$g_1^2\,\lambda_{HS_2}^{(1)}$	$-\frac{51}{2}$
$\lambda_1\,\lambda_{HS_2}^{(1)}$	6
$\left(\lambda_{HS_2}^{(1)}\right)^2$	4
$\lambda_{HS_2}^{(1)}\,\lambda_{S_2}^{(1)}$	8
$\lambda_{12}^{(1)}\,\lambda_{HS_1}^{(1)}$	6
g_2^4	6
g_1^4	12
$\left(\lambda_{HS_2}^{(A)}\right)^2$	2
$\lambda_{HS_2}^{(1)}\,\lambda_{S_2}^{(A)}$	4
$\lambda_{HS_2}^{(1)}\,T$	2
$\lambda_{HS_1}^{(1)}\,\lambda_{12}^{(\times)}$	4

$\beta_{\lambda_{12}^{(1)}}$

Run block 1: T3-A-m4–T3-B-m1

$\mathcal{S}$	$\mathrm{T3-A-m4}\ (\alpha=-4)$	$\mathrm{T3-A-m2}\ (\alpha=-2)$	$\mathrm{T3-A-p0}\ (\alpha=0)$	$\mathrm{T3-B-m3}\ (\alpha=$
$\lvert\lambda_5\rvert^2$	4	4	4	3
$g_2^2\,\lambda_{12}^{(1)}$	-12	-12	-12	-9
$g_1^2\,\lambda_{12}^{(1)}$	-30	-6	-6	-15
$\left(\lambda_{12}^{(1)}\right)^2$	4	4	4	4
$\lambda_{12}^{(1)}\,\lambda_{S_1}^{(1)}$	4	4	4	6
$\lambda_{12}^{(1)}\,\lambda_{S_2}^{(1)}$	8	8	8	6
$\lambda_{HS_1}^{(1)}\,\lambda_{HS_2}^{(1)}$	4	4	4	4
g_2^4	0	0	0	$\frac{9}{4}$
g_1^4	48	0	0	$\frac{27}{4}$
$\lambda_{12}^{(1)}\,\lambda_{S_2}^{(A)}$	4	4	4	0
$\left(\lambda_{12}^{(A)}\right)^2$	0	0	0	$\frac{3}{4}$
$\lambda_{12}^{(1)}\,\lambda_{S_1}^{(A)}$	0	0	0	0
$\left(\lambda_{12}^{(\times)}\right)^2$	0	0	0	0
$\lambda_{12}^{(\times)}\,\lambda_{S_1}^{(A)}$	0	0	0	0
$\lambda_{12}^{(\times)}\,\lambda_{S_2}^{(A)}$	0	0	0	0

Run block 2: T3-B-p1–T3-D-m2

$\mathcal{S}$	$\mathrm{T3-B-p1}\ (\alpha=1)$	$\mathrm{T3-C-m3}\ (\alpha=-3)$	$\mathrm{T3-C-m1}\ (\alpha=-1)$	$\mathrm{T3-C-p1}\ (\alpha=$
$\lvert\lambda_5\rvert^2$	3	3	3	3
$g_2^2\,\lambda_{12}^{(1)}$	-9	-9	-9	-9
$g_1^2\,\lambda_{12}^{(1)}$	-15	-15	-3	-15
$\left(\lambda_{12}^{(1)}\right)^2$	4	4	4	4
$\lambda_{12}^{(1)}\,\lambda_{S_1}^{(1)}$	6	6	6	6
$\lambda_{12}^{(1)}\,\lambda_{S_2}^{(1)}$	6	6	6	6
$\lambda_{HS_1}^{(1)}\,\lambda_{HS_2}^{(1)}$	4	4	4	4
g_2^4	$\frac{9}{4}$	$\frac{9}{4}$	$\frac{9}{4}$	$\frac{9}{4}$
g_1^4	$\frac{27}{4}$	$\frac{27}{4}$	$\frac{3}{4}$	$\frac{27}{4}$
$\lambda_{12}^{(1)}\,\lambda_{S_2}^{(A)}$	0	0	0	0
$\left(\lambda_{12}^{(A)}\right)^2$	$\frac{3}{4}$	$\frac{3}{4}$	$\frac{3}{4}$	$\frac{3}{4}$
$\lambda_{12}^{(1)}\,\lambda_{S_1}^{(A)}$	0	0	0	0
$\left(\lambda_{12}^{(\times)}\right)^2$	0	0	0	0
$\lambda_{12}^{(\times)}\,\lambda_{S_1}^{(A)}$	0	0	0	0
$\lambda_{12}^{(\times)}\,\lambda_{S_2}^{(A)}$	0	0	0	0

Run block 3: T3-D-p0–T3-E-p0

$\mathcal{S}$	$\mathrm{T3-D-p0}\ (\alpha=0)$	$\mathrm{T3-D-p2}\ (\alpha=2)$	$\mathrm{T3-E-m4}\ (\alpha=-4)$	$\mathrm{T3-E-m2}\ (\alpha=-$
$\lvert\lambda_5\rvert^2$	4	4	$\frac{4}{3}$	$\frac{4}{3}$
$g_2^2\,\lambda_{12}^{(1)}$	-12	-12	-24	-24
$g_1^2\,\lambda_{12}^{(1)}$	-6	-30	-30	-6
$\left(\lambda_{12}^{(1)}\right)^2$	4	4	4	4
$\lambda_{12}^{(1)}\,\lambda_{S_1}^{(1)}$	8	8	8	8
$\lambda_{12}^{(1)}\,\lambda_{S_2}^{(1)}$	4	4	8	8
$\lambda_{HS_1}^{(1)}\,\lambda_{HS_2}^{(1)}$	4	4	4	4

g_2^4	0	0	12	12
g_1^4	0	48	48	0
$\lambda_{12}^{(1)}\,\lambda_{S_2}^{(A)}$	0	0	4	4
$\left(\lambda_{12}^{(A)}\right)^2$	0	0	4	4
$\lambda_{12}^{(1)}\,\lambda_{S_1}^{(A)}$	4	4	4	4
$\left(\lambda_{12}^{(\times)}\right)^2$	0	0	4	4
$\lambda_{12}^{(\times)}\,\lambda_{S_1}^{(A)}$	0	0	4	4
$\lambda_{12}^{(\times)}\,\lambda_{S_2}^{(A)}$	0	0	4	4

Run block 4: T3-E-p2–T3-E-p2

$\mathcal{S}$	$\mathrm{T3-E-p2}\ (\alpha=2)$
$\lvert\lambda_5\rvert^2$	$\frac{4}{3}$
$g_2^2\,\lambda_{12}^{(1)}$	-24
$g_1^2\,\lambda_{12}^{(1)}$	-30
$\left(\lambda_{12}^{(1)}\right)^2$	4
$\lambda_{12}^{(1)}\,\lambda_{S_1}^{(1)}$	8
$\lambda_{12}^{(1)}\,\lambda_{S_2}^{(1)}$	8
$\lambda_{HS_1}^{(1)}\,\lambda_{HS_2}^{(1)}$	4
g_2^4	12
g_1^4	48
$\lambda_{12}^{(1)}\,\lambda_{S_2}^{(A)}$	4
$\left(\lambda_{12}^{(A)}\right)^2$	4
$\lambda_{12}^{(1)}\,\lambda_{S_1}^{(A)}$	4
$\left(\lambda_{12}^{(\times)}\right)^2$	4
$\lambda_{12}^{(\times)}\,\lambda_{S_1}^{(A)}$	4
$\lambda_{12}^{(\times)}\,\lambda_{S_2}^{(A)}$	4

β_{λ_5}

Run block 1: T3-A-m4–T3-B-m1

$\mathcal{S}$	$\mathrm{T3-A-m4}\ (\alpha=-4)$	$\mathrm{T3-A-m2}\ (\alpha=-2)$	$\mathrm{T3-A-p0}\ (\alpha=0)$	$\mathrm{T3-B-m3}\ (\alpha=$
$\lambda_5\,\lambda_1$	2	2	2	2
$\lambda_5\,\lambda_{HS_1}^{(1)}$	4	4	4	4
$\lambda_5\,\lambda_{HS_2}^{(1)}$	4	4	4	4
$\lambda_5\,\lambda_{12}^{(1)}$	2	2	2	2
$g_2^2\,\lambda_5$	$-\frac{21}{2}$	$-\frac{21}{2}$	$-\frac{21}{2}$	-9
$g_1^2\,\lambda_5$	$-\frac{33}{2}$	$-\frac{9}{2}$	$-\frac{9}{2}$	-9
$\lambda_5\,\lambda_{HS_2}^{(A)}$	4	4	4	2
$\lambda_5\,T$	2	2	2	2
$\lambda_5\,\lambda_{HS_1}^{(A)}$	0	0	0	-2
$\lambda_5\,\lambda_{12}^{(A)}$	0	0	0	$-\frac{1}{2}$
$\lambda_5\,\lambda_{12}^{(\times)}$	0	0	0	0

Run block 2: T3-B-p1–T3-D-m2

$\mathcal{S}$	$\mathrm{T3-B-p1}\ (\alpha=1)$	$\mathrm{T3-C-m3}\ (\alpha=-3)$	$\mathrm{T3-C-m1}\ (\alpha=-1)$	$\mathrm{T3-C-p1}\ (\alpha=$
$\lambda_5\,\lambda_1$	2	2	2	2
$\lambda_5\,\lambda_{HS_1}^{(1)}$	4	4	4	4

$\lambda_5\,\lambda_{HS_2}^{(1)}$	4	4	4	4
$\lambda_5\,\lambda_{12}^{(1)}$	2	2	2	2
$g_2^2\,\lambda_5$	-9	-9	-9	-9
$g_1^2\,\lambda_5$	-9	-9	-3	-9
$\lambda_5\,\lambda_{HS_2}^{(A)}$	2	2	2	2
$\lambda_5\,T$	2	2	2	2
$\lambda_5\,\lambda_{HS_1}^{(A)}$	-2	-2	-2	-2
$\lambda_5\,\lambda_{12}^{(A)}$	$-\frac{1}{2}$	$-\frac{1}{2}$	$-\frac{1}{2}$	$-\frac{1}{2}$
$\lambda_5\,\lambda_{12}^{(\times)}$	0	0	0	0

Run block 3: T3-D-p0–T3-E-p0

$\mathcal{S}$	$\mathrm{T3-D-p0}\ (\alpha=0)$	$\mathrm{T3-D-p2}\ (\alpha=2)$	$\mathrm{T3-E-m4}\ (\alpha=-4)$	$\mathrm{T3-E-m2}\ (\alpha=-$
$\lambda_5\,\lambda_1$	2	2	2	2
$\lambda_5\,\lambda_{HS_1}^{(1)}$	4	4	4	4
$\lambda_5\,\lambda_{HS_2}^{(1)}$	4	4	4	4
$\lambda_5\,\lambda_{12}^{(1)}$	2	2	2	2
$g_2^2\,\lambda_5$	$-\frac{21}{2}$	$-\frac{21}{2}$	$-\frac{33}{2}$	$-\frac{33}{2}$
$g_1^2\,\lambda_5$	$-\frac{9}{2}$	$-\frac{33}{2}$	$-\frac{33}{2}$	$-\frac{9}{2}$
$\lambda_5\,\lambda_{HS_2}^{(A)}$	0	0	2	2
$\lambda_5\,T$	2	2	2	2
$\lambda_5\,\lambda_{HS_1}^{(A)}$	-4	-4	-2	-2
$\lambda_5\,\lambda_{12}^{(A)}$	0	0	2	2
$\lambda_5\,\lambda_{12}^{(\times)}$	0	0	-2	-2

Run block 4: T3-E-p2–T3-E-p2

$\mathcal{S}$	$\mathrm{T3-E-p2}\ (\alpha=2)$
$\lambda_5\,\lambda_1$	2
$\lambda_5\,\lambda_{HS_1}^{(1)}$	4
$\lambda_5\,\lambda_{HS_2}^{(1)}$	4
$\lambda_5\,\lambda_{12}^{(1)}$	2
$g_2^2\,\lambda_5$	$-\frac{33}{2}$
$g_1^2\,\lambda_5$	$-\frac{33}{2}$
$\lambda_5\,\lambda_{HS_2}^{(A)}$	2
$\lambda_5\,T$	2
$\lambda_5\,\lambda_{HS_1}^{(A)}$	-2
$\lambda_5\,\lambda_{12}^{(A)}$	2
$\lambda_5\,\lambda_{12}^{(\times)}$	-2

Additional scalar-coupling beta functions

$\beta_{\lambda_{HS_1}^{(A)}}$

Run block 1: T3-A-m4–T3-B-m1

$\mathcal{S}$	$\mathrm{T3-A-m4}\ (\alpha=-4)$	$\mathrm{T3-A-m2}\ (\alpha=-2)$	$\mathrm{T3-A-p0}\ (\alpha=0)$	$\mathrm{T3-B-m3}\ (\alpha=$
$g_1^2 g_2^2$	0	0	0	-18

Run block 2: T3-B-p1–T3-D-m2

$\mathcal{S}$	T3 − B − p1 ($\alpha = 1$)	T3 − C − m3 ($\alpha = -3$)	T3 − C − m1 ($\alpha = -1$)	T3 − C − p1 ($\alpha =$
$g_1^2 g_2^2$	6	−18	−6	6

Run block 3: T3-D-p0–T3-E-p0

$\mathcal{S}$	T3 − D − p0 ($\alpha = 0$)	T3 − D − p2 ($\alpha = 2$)	T3 − E − m4 ($\alpha = -4$)	T3 − E − m2 ($\alpha = -$
$g_1^2 g_2^2$	0	12	−24	−12

Run block 4: T3-E-p2–T3-E-p2

$\mathcal{S}$	T3 − E − p2 ($\alpha = 2$)
$g_1^2 g_2^2$	12

$\beta_{\lambda_{HS_2}^{(A)}}$

Run block 1: T3-A-m4–T3-B-m1

$\mathcal{S}$	T3 − A − m4 ($\alpha = -4$)	T3 − A − m2 ($\alpha = -2$)	T3 − A − p0 ($\alpha = 0$)	T3 − B − m3 ($\alpha =$
$g_1^2 g_2^2$	−12	0	12	−6

Run block 2: T3-B-p1–T3-D-m2

$\mathcal{S}$	T3 − B − p1 ($\alpha = 1$)	T3 − C − m3 ($\alpha = -3$)	T3 − C − m1 ($\alpha = -1$)	T3 − C − p1 ($\alpha =$
$g_1^2 g_2^2$	18	−6	6	18

Run block 3: T3-D-p0–T3-E-p0

$\mathcal{S}$	T3 − D − p0 ($\alpha = 0$)	T3 − D − p2 ($\alpha = 2$)	T3 − E − m4 ($\alpha = -4$)	T3 − E − m2 ($\alpha = -$
$g_1^2 g_2^2$	0	0	−12	0

Run block 4: T3-E-p2–T3-E-p2

$\mathcal{S}$	T3 − E − p2 ($\alpha = 2$)
$g_1^2 g_2^2$	24

$\beta_{\lambda_{12}^{(A)}}$

Run block 1: T3-A-m4–T3-B-m1

$\mathcal{S}$	T3 − A − m4 ($\alpha = -4$)	T3 − A − m2 ($\alpha = -2$)	T3 − A − p0 ($\alpha = 0$)	T3 − B − m3 ($\alpha =$
$g_1^2 g_2^2$	0	0	0	18

Run block 2: T3-B-p1–T3-D-m2

$\mathcal{S}$	T3 − B − p1 ($\alpha = 1$)	T3 − C − m3 ($\alpha = -3$)	T3 − C − m1 ($\alpha = -1$)	T3 − C − p1 ($\alpha =$
$g_1^2 g_2^2$	18	18	−6	18

Run block 3: T3-D-p0–T3-E-p0

$\mathcal{S}$	T3 − D − p0 ($\alpha = 0$)	T3 − D − p2 ($\alpha = 2$)	T3 − E − m4 ($\alpha = -4$)	T3 − E − m2 ($\alpha = -$
$g_1^2 g_2^2$	0	0	48	0

Run block 4: T3-E-p2–T3-E-p2

$\mathcal{S}$	T3 − E − p2 ($\alpha = 2$)
$g_1^2 g_2^2$	48

$\beta_{\lambda_{12}^{(\times)}}$

Run block 1: T3-A-m4–T3-B-m1

$\mathcal{S}$	T3 − A − m4 ($\alpha = -4$)	T3 − A − m2 ($\alpha = -2$)	T3 − A − p0 ($\alpha = 0$)	T3 − B − m3 ($\alpha =$
g_2^4	0	0	0	0
$\left(\lambda_{12}^{(A)}\right)^2$	0	0	0	0
$\lambda_{12}^{(1)}\,\lambda_{12}^{(\times)}$	0	0	0	0
$\left(\lambda_{12}^{(\times)}\right)^2$	0	0	0	0
$\lambda_{12}^{(\times)}\,\lambda_{S_1}^{(A)}$	0	0	0	0
$\lambda_{12}^{(\times)}\,\lambda_{S_2}^{(A)}$	0	0	0	0

Run block 2: T3-B-p1–T3-D-m2

$\mathcal{S}$	T3 − B − p1 ($\alpha = 1$)	T3 − C − m3 ($\alpha = -3$)	T3 − C − m1 ($\alpha = -1$)	T3 − C − p1 ($\alpha =$
g_2^4	0	0	0	0
$\left(\lambda_{12}^{(A)}\right)^2$	0	0	0	0
$\lambda_{12}^{(1)}\,\lambda_{12}^{(\times)}$	0	0	0	0
$\left(\lambda_{12}^{(\times)}\right)^2$	0	0	0	0
$\lambda_{12}^{(\times)}\,\lambda_{S_1}^{(A)}$	0	0	0	0
$\lambda_{12}^{(\times)}\,\lambda_{S_2}^{(A)}$	0	0	0	0

Run block 3: T3-D-p0–T3-E-p0

$\mathcal{S}$	T3 − D − p0 ($\alpha = 0$)	T3 − D − p2 ($\alpha = 2$)	T3 − E − m4 ($\alpha = -4$)	T3 − E − m2 ($\alpha = -$
g_2^4	0	0	6	6
$\left(\lambda_{12}^{(A)}\right)^2$	0	0	2	2
$\lambda_{12}^{(1)}\,\lambda_{12}^{(\times)}$	0	0	8	8
$\left(\lambda_{12}^{(\times)}\right)^2$	0	0	10	10
$\lambda_{12}^{(\times)}\,\lambda_{S_1}^{(A)}$	0	0	−2	−2
$\lambda_{12}^{(\times)}\,\lambda_{S_2}^{(A)}$	0	0	−2	−2

Run block 4: T3-E-p2–T3-E-p2

$\mathcal{S}$	T3 − E − p2 ($\alpha = 2$)
g_2^4	6
$\left(\lambda_{12}^{(A)}\right)^2$	2
$\lambda_{12}^{(1)}\,\lambda_{12}^{(\times)}$	8
$\left(\lambda_{12}^{(\times)}\right)^2$	10

$\lambda_{12}^{(\times)}\,\lambda_{S_1}^{(A)}$	-2
$\lambda_{12}^{(\times)}\,\lambda_{S_2}^{(A)}$	-2

Full Weinberg-coefficient RGE in EFT1

Writing the canonical mixed LLSS coefficient as C_{12}, the coupled one-loop Weinberg equation used in the F-first EFT is

$$16\pi^2\frac{dC_5}{d\ln\mu} = (2\lambda_1 - 3g_2^2 + 2T)C_5 \tag{1}$$

$$-\frac{3}{2}\left[(Y_e^\dagger Y_e)^T C_5 + C_5(Y_e^\dagger Y_e)\right] \tag{2}$$

$$+ R_W\,\lambda_5\,C_{12}. \tag{3}$$

Note: $\frac{7}{2}$ thing isnt there if we integrate out our fermion F so its not here

Direct $C_{12} \to C_5$ recoupling factor

For each representation the source term is

$$16\pi^2\,\beta_{C_5} \supset R_W\,\lambda_5\,C_{12}.$$

$\mathcal{S}$	T3 − A − m4 ($\alpha=-4$)	T3 − A − m2 ($\alpha=-2$)	T3 − A − p0 ($\alpha=0$)	T3 − B − m3 ($\alpha=-3$)	T3 − B − m1 ($\alpha=-1$)
R_W	$\frac{4\sqrt{3}}{3}$	$\frac{4\sqrt{3}}{3}$	$\frac{4\sqrt{3}}{3}$	$\frac{4\sqrt{3}}{3}$	$\frac{4\sqrt{3}}{3}$

$\mathcal{S}$	T3 − B − p1 ($\alpha=1$)	T3 − C − m3 ($\alpha=-3$)	T3 − C − m1 ($\alpha=-1$)	T3 − C − p1 ($\alpha=1$)	T3 − D − m2 ($\alpha=-2$)
R_W	$\frac{4\sqrt{3}}{3}$	$-\frac{4}{3}$	$-\frac{4}{3}$	$-\frac{4}{3}$	$\frac{4\sqrt{3}}{3}$

$\mathcal{S}$	T3 − D − p0 ($\alpha=0$)	T3 − D − p2 ($\alpha=2$)	T3 − E − m4 ($\alpha=-4$)	T3 − E − m2 ($\alpha=-2$)	T3 − E − p0 ($\alpha=0$)
R_W	$\frac{4\sqrt{3}}{3}$	$\frac{4\sqrt{3}}{3}$	$-\frac{4\sqrt{6}}{3}$	$-\frac{4\sqrt{6}}{3}$	$-\frac{4\sqrt{6}}{3}$

$\mathcal{S}$	T3 − E − p2 ($\alpha=2$)
R_W	$-\frac{4\sqrt{6}}{3}$

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T3 − A − m4 ($\alpha=-4$)

1 independent complex matched structure displayed.

complex matched operator	matched prefactor	$SU(2)_L$ invariant
$(L_i^T C L_j)\,S_1^\dagger\,S_2 + \text{h.c.}$	$\frac{\overline{y_1}\,\overline{y_2}}{M_F}$	$\mathcal{I}_{LLS_1S_2}$

T3 − A − m2 ($\alpha=-2$)

1 independent complex matched structure displayed.

complex matched operator	matched prefactor	$SU(2)_L$ invariant
$(L_i^T C L_j)\,S_1^\dagger\,S_2 + \text{h.c.}$	$\frac{\overline{y_1}\,\overline{y_2}}{M_F}$	$\mathcal{I}_{LLS_1S_2}$

T3 − A − p0 ($\alpha = 0$)

1 independent complex matched structure displayed.

complex matched operator	matched prefactor	$SU(2)_L$ invariant
$(L_i^T C L_j)\, S_1^\dagger\, S_2 + \text{h.c.}$	$\frac{\overline{y_1 y_2}}{M_F}$	$\mathcal{I}_{LLS_1S_2}$

T3 − B − m3 ($\alpha = -3$)

1 independent complex matched structure displayed.

complex matched operator	matched prefactor	$SU(2)_L$ invariant
$(L_i^T C L_j)\, S_1^\dagger\, S_2 + \text{h.c.}$	$\frac{\overline{y_1 y_2}}{M_F}$	$\mathcal{C}_{LS_1F}\,\mathcal{C}_{LS_2F}$

T3 − B − m1 ($\alpha = -1$)

3 independent complex matched structures displayed.

complex matched operator	matched prefactor	$SU(2)_L$ invariant
$(L_i^T C L_j)\, S_1^\dagger\, S_1^\dagger + \text{h.c.}$	$\frac{\overline{y_1}^2}{2M_F}$	$\mathcal{C}_{LS_1F}$
$(L_i^T C L_j)\, S_1^\dagger\, S_2 + \text{h.c.}$	$\frac{\overline{y_1 y_2}}{M_F}$	$\mathcal{C}_{LS_1F}\,\mathcal{C}_{LS_2F}$
$(L_i^T C L_j)\, S_2\, S_2 + \text{h.c.}$	$\frac{\overline{y_2}^2}{2M_F}$	$\mathcal{C}_{LS_2F}$

T3 − B − p1 ($\alpha = 1$)

1 independent complex matched structure displayed.

complex matched operator	matched prefactor	$SU(2)_L$ invariant
$(L_i^T C L_j)\, S_1^\dagger\, S_2 + \text{h.c.}$	$\frac{\overline{y_1 y_2}}{M_F}$	$\mathcal{C}_{LS_1F}\,\mathcal{C}_{LS_2F}$

T3 − C − m3 ($\alpha = -3$)

2 independent complex matched structures displayed.

complex matched operator	matched prefactor	$SU(2)_L$ invariant
$(L_i^T C L_j)\, S_1^\dagger\, S_2 + \text{h.c.}$	$-\frac{\sqrt{3}\overline{y_1 y_2}}{2M_F}$	ϵ
$(L_i^T C L_j)\, S_1^\dagger\, S_2 + \text{h.c.}$	$-\frac{\overline{y_1 y_2}}{2M_F}$	$\mathcal{I}_{LLS_1S_2}$

T3 − C − m1 ($\alpha = -1$)

4 independent complex matched structures displayed.

complex matched operator	matched prefactor	$SU(2)_L$ invariant
$(L_i^T C L_j)\, S_1^\dagger\, S_1^\dagger + \text{h.c.}$	$\frac{\overline{y_1}^2}{2M_F}$	cg2
$(L_i^T C L_j)\, S_1^\dagger\, S_2 + \text{h.c.}$	$-\frac{\sqrt{3}\overline{y_1 y_2}}{2M_F}$	ϵ
$(L_i^T C L_j)\, S_1^\dagger\, S_2 + \text{h.c.}$	$-\frac{\overline{y_1 y_2}}{2M_F}$	$\mathcal{I}_{LLS_1S_2}$
$(L_i^T C L_j)\, S_2\, S_2 + \text{h.c.}$	$\frac{\overline{y_2}^2}{2M_F}$	cg1

T3 − C − p1 ($\alpha = 1$)

2 independent complex matched structures displayed.

complex matched operator	matched prefactor	$SU(2)_L$ invariant
$(L_i^T C L_j)\, S_1^\dagger\, S_2 + \text{h.c.}$	$-\frac{\sqrt{3}\overline{y_1 y_2}}{2M_F}$	ϵ
$(L_i^T C L_j)\, S_1^\dagger\, S_2 + \text{h.c.}$	$-\frac{\overline{y_1 y_2}}{2M_F}$	$\mathcal{I}_{LLS_1S_2}$

$\mathrm{T3-D-m2}$ $(\alpha = -2)$

1 independent complex matched structure displayed.

complex matched operator	matched prefactor	$SU(2)_L$ invariant
$(L_i^T C L_j)\, S_1^\dagger\, S_2 + \text{h.c.}$	$\frac{\overline{y_1 y_2}}{M_F}$	$\mathcal{I}_{LLS_1S_2}$

$\mathrm{T3-D-p0}$ $(\alpha = 0)$

1 independent complex matched structure displayed.

complex matched operator	matched prefactor	$SU(2)_L$ invariant
$(L_i^T C L_j)\, S_1^\dagger\, S_2 + \text{h.c.}$	$\frac{\overline{y_1 y_2}}{M_F}$	$\mathcal{I}_{LLS_1S_2}$

$\mathrm{T3-D-p2}$ $(\alpha = 2)$

1 independent complex matched structure displayed.

complex matched operator	matched prefactor	$SU(2)_L$ invariant
$(L_i^T C L_j)\, S_1^\dagger\, S_2 + \text{h.c.}$	$\frac{\overline{y_1 y_2}}{M_F}$	$\mathcal{I}_{LLS_1S_2}$

$\mathrm{T3-E-m4}$ $(\alpha = -4)$

2 independent complex matched structures displayed.

complex matched operator	matched prefactor	$SU(2)_L$ invariant
$(L_i^T C L_j)\, S_1^\dagger\, S_2 + \text{h.c.}$	$\frac{\sqrt{3}\overline{y_1 y_2}}{3M_F}$	ϵ
$(L_i^T C L_j)\, S_1^\dagger\, S_2 + \text{h.c.}$	$\frac{\sqrt{6}\overline{y_1 y_2}}{3M_F}$	$\mathcal{I}_{LLS_1S_2}$

$\mathrm{T3-E-m2}$ $(\alpha = -2)$

2 independent complex matched structures displayed.

complex matched operator	matched prefactor	$SU(2)_L$ invariant
$(L_i^T C L_j)\, S_1^\dagger\, S_2 + \text{h.c.}$	$\frac{\sqrt{3}\overline{y_1 y_2}}{3M_F}$	ϵ
$(L_i^T C L_j)\, S_1^\dagger\, S_2 + \text{h.c.}$	$\frac{\sqrt{6}\overline{y_1 y_2}}{3M_F}$	$\mathcal{I}_{LLS_1S_2}$

$\mathrm{T3-E-p0}$ $(\alpha = 0)$

2 independent complex matched structures displayed.

complex matched operator	matched prefactor	$SU(2)_L$ invariant
$(L_i^T C L_j)\, S_1^\dagger\, S_2 + \text{h.c.}$	$\frac{\sqrt{3}\overline{y_1 y_2}}{3M_F}$	ϵ
$(L_i^T C L_j)\, S_1^\dagger\, S_2 + \text{h.c.}$	$\frac{\sqrt{6}\overline{y_1 y_2}}{3M_F}$	$\mathcal{I}_{LLS_1S_2}$

T3 − E − p2 ($\alpha = 2$)

2 independent complex matched structures displayed.

complex matched operator	matched prefactor	$SU(2)_L$ invariant
$(L_i^T C L_j)\, S_1^\dagger\, S_2 + \text{h.c.}$	$\frac{\sqrt{3}\overline{y_1 y_2}}{3M_F}$	ϵ
$(L_i^T C L_j)\, S_1^\dagger\, S_2 + \text{h.c.}$	$\frac{\sqrt{6}\overline{y_1 y_2}}{3M_F}$	$\mathcal{I}_{LLS_1S_2}$

Next threshold

Further matching depends on the next field or fields selected by the threshold plan. Once all T3 fields are removed, their representation dependence is carried by the matched C_5 boundary condition and the subsequent SMEFT running is universal.